



YEHIA WAEL SAAD

SENIOR FULL-STACK ENGINEER & INTERACTIVE SYSTEMS ARCHITECT

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Location: Cairo, Egypt (Remote)

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CORE TECHNICAL ARSENAL

Languages

TypeScript, JavaScript (ESNext), Python, Go (Golang), SQL, HTML5/CSS3

Frontend Architecture

React 19, Next.js, Tailwind CSS, Vite, Redux Toolkit, Zustand, Three.js, WebGL, Canvas 2D API

Backend & Services

Node.js, Express, Fastify, Microservices, RESTful APIs, WebSockets, JWT/OAuth2, gRPC

Databases & Storage

PostgreSQL, LibSQL / Turso, SQLite, Redis, MongoDB, Prisma ORM, Query Optimization

Cloud & DevOps

Docker, Kubernetes, AWS, Vercel, CI/CD, GitHub Actions, Linux Systems, Nginx

KEY COMPETENCIES

- Distributed Systems Design
- Real-Time WebSocket Telemetry
- High-Concurrency Architecture
- Interactive WebGL & 3D Shaders
- Database Tuning & Indexing
- Automated CI/CD Pipelines

LANGUAGES

English: Fluent / Professional (C1)
Arabic: Native

EXECUTIVE SUMMARY

Accomplished Senior Full-Stack Engineer and Systems Architect with over 5+ years of specialized engineering experience forging distributed high-throughput backends, fault-tolerant microservices, and cutting-edge interactive web experiences. Proven track record scaling low-latency API architectures handling 120M+ monthly requests, architecting real-time WebSocket telemetry pipelines, and engineering high-performance WebGL graphics applications.

PROFESSIONAL EXPERIENCE

Lead Architect & Senior Full-Stack Engineer

2023 - Present

Apex Cyber Systems | Cairo / Remote

- Architected and deployed distributed event-driven microservices handling 120M+ monthly transactions with 99.99% system availability.
- Engineered real-time bidirectional WebSocket telemetry pipeline with sub-15ms packet latency, reducing data lag by 45%.
- Spearheaded database partitioning, indexing strategies, and multi-tier Redis caching, reducing server load by 55%.
- Mentored squad of 8 software engineers, instituted code quality gates, automated testing, and zero-downtime CI/CD workflows.

Tech Stack: TypeScript, Node.js, Go, PostgreSQL, Redis, Docker, WebSockets, Tailwind CSS

Senior Full-Stack Engineer

2021 - 2023

Nexus Interactive | Remote Worldwide

- Engineered reactive modular web applications and enterprise dashboards using React 19, Next.js, and TypeScript.
- Optimized complex SQL and NoSQL aggregation queries, trimming P95 query execution times from 420ms to 85ms.
- Implemented 2D/3D procedural WebGL canvas rendering tools for interactive data visualization with 60 FPS performance.

Tech Stack: React, TypeScript, Next.js, Tailwind CSS, Express, MongoDB, WebGL, Jest

EDUCATION

B.Sc. in Computer Science & Software Engineering

Cairo University (2017 - 2021)

Honors: Excellent / First Class Honors
Focused on Distributed Algorithms, Advanced Operating Systems, Cryptography, and Database Systems Engineering.

Advanced Cloud Architecture & Systems Design

Institute of Software Systems (2021 - 2022)

Honors: Certified Systems Architect
Specialization in high-concurrency microservices, multi-region database failover, and zero-trust security topologies.

KEY METRICS

Production Deployments

28+ Systems

Industry Experience

5+ Years

Uptime Reliability

99.99% SLA

Code Quality Benchmark

A+ / Clean Arch

PAST EXPERIENCE (CONTINUED)

Full-Stack Web & Software Developer

2019 - 2021

Kinetic Software Labs | Cairo, Egypt

- Developed and deployed mission-critical RESTful APIs and payment processing integrations for consumer-facing portals.
- Engineered client portals with dynamic responsive layouts, cross-browser consistency, and rigorous unit/integration test coverage.

Tech Stack: JavaScript, Node.js, Express, PostgreSQL, Redis, REST APIs, HTML5/CSS3

FLAGSHIP ENGINEERING PROJECTS

AI Cyber Sentinel [Security & Artificial Intelligence]

Autonomous neural threat intelligence platform performing real-time packet inspection, automated intrusion detection, and instant threat neutralization telemetry.

Key Architecture: Distributed worker pool processing 25,000 security events per second.

Technologies: React 19, TypeScript, Node.js, TensorFlow, WebSockets, Tailwind CSS

Quantum Flow Canvas [Interactive Graphics & 3D WebGL]

High-performance procedural graphics editor featuring node-based compositing, custom WebGL fragment shader execution, and hardware-accelerated canvas exporting.

Key Architecture: Sub-16ms render loop maintaining consistent 60 FPS under complex particle simulations.

Technologies: TypeScript, HTML5 Canvas, WebGL 2.0, Vite, Tailwind CSS

Hyper-Scale FinTech Core [Distributed Ledger & FinTech]

High-concurrency distributed financial transaction ledger engine with ACID guarantees, double-entry bookkeeping, and real-time reconciliation audits.

Key Architecture: Benchmarked at 50,000+ financial transactions per second with zero drift.

Technologies: Go (Golang), Node.js, PostgreSQL, Redis, Docker, Microservices

Matrix Game Engine & ECS [Game Tech & Distributed Multi-Player]

Web-based high-concurrency Entity Component System engine for real-time multiplayer simulation with server-authoritative state reconciliation.

Key Architecture: Client-side prediction and lag compensation for up to 64 concurrent players.

Technologies: TypeScript, WebAssembly, WebSockets, Three.js, Node.js